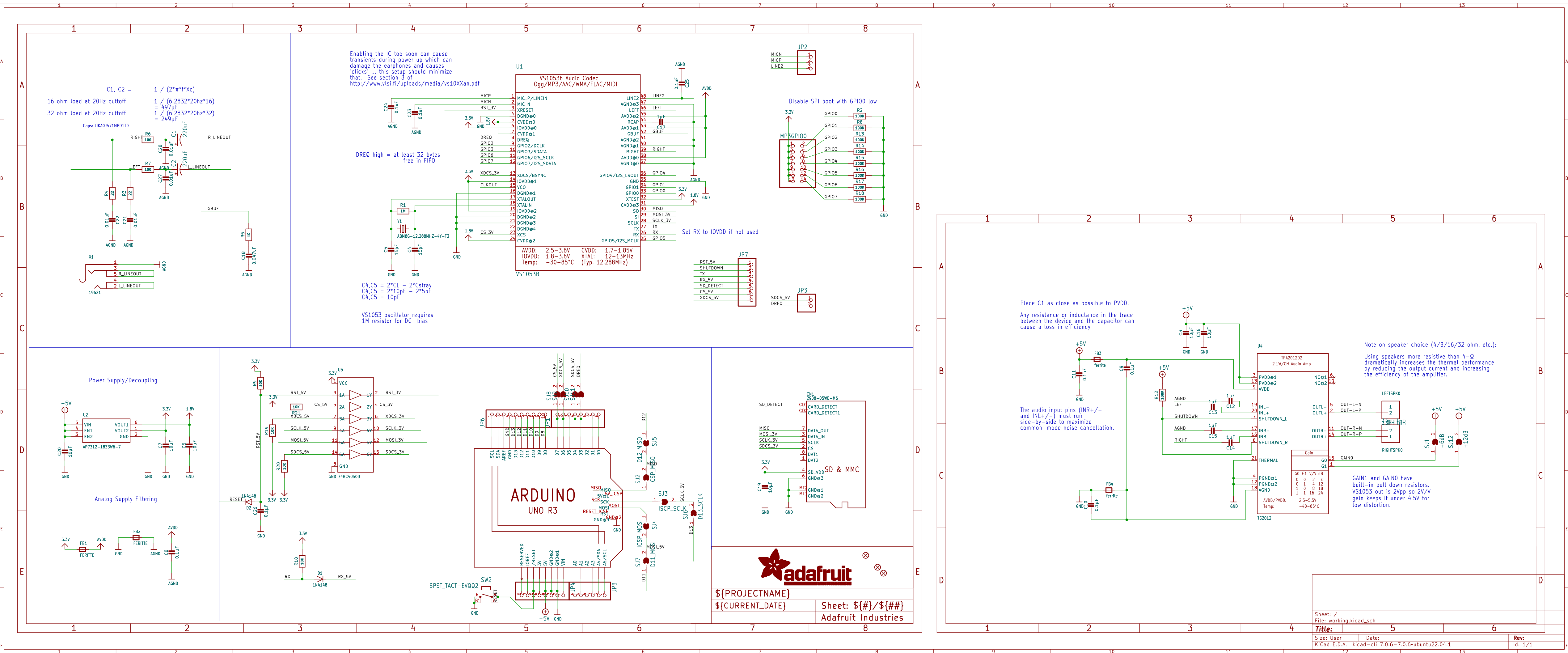




$C1, C2 = 1 / (2 * \pi * f * X_c)$
 $16 \text{ ohm load at } 20\text{Hz cutoff} = 1 / (6.2832 * 20\text{Hz} * 16) = 497\mu\text{F}$
 $32 \text{ ohm load at } 20\text{Hz cutoff} = 1 / (6.2832 * 20\text{Hz} * 32) = 249\mu\text{F}$
Caps: UKA0J471MPD1TD

Enabling the IC too soon can cause transients during power up which can damage the earphones and causes 'clicks' ... this setup should minimize that. See section 8 of <http://www.vlsi.fi/uploads/media/vs10XXan.pdf>

DREQ high = at least 32 bytes free in FIFO

VS1053 oscillator requires 1M resistor for DC bias

$C4, C5 = 2 * C_L - 2 * C_{stray}$
 $C4, C5 = 2 * 10\text{pF} - 2 * 5\text{pF}$
 $C4, C5 = 10\text{pF}$

Disable SPI boot with GPIO0 low

Set RX to IOVDD if not used

Place C1 as close as possible to PVDD.

Any resistance or inductance in the trace between the device and the capacitor can cause a loss in efficiency

The audio input pins (INR+/- and INL+/-) must run side-by-side to maximize common-mode noise cancellation.

Note on speaker choice (4/8/16/32 ohm, etc.):
Using speakers more resistive than 4-Ω dramatically increases the thermal performance by reducing the output current and increasing the efficiency of the amplifier.

GAIN1 and GAIN0 have built-in pull down resistors. VS1053 out is 2Vpp so 2V/V gain keeps it under 4.5V for low distortion.



{PROJECTNAME}
{CURRENT_DATE} Sheet: {#}/{##}
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Sheet: /		File: working.kicad_sch	
Title:			
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